

Speed: $v = s/t$

Average speed, unit conversion and distance–time graphs

Physics · 2A · 11 September 2026

Who arrives first?

(slide 2)

You and a friend leave Piazza Maggiore together. One walks, one cycles, one drives.

Before moving on. Who arrives first, and how do we know without a stopwatch?

It depends on two things: how much road there is to cover, and how long it takes. That is all we need to compare the three of you.

Average speed

(slide 3)

What you have just described has a name: **average speed**.

$$v = \frac{s}{t}$$

s is the distance (m), t is the time (s), v is the speed (m/s).

Common unit: km/h.

km/h $\rightarrow$ m/s: divide by 3.6 m/s $\rightarrow$ km/h: multiply by 3.6

In the SI, s is measured in metres and t in seconds, so v is in metres per second. In everyday life you almost always meet it in km/h: the speedometer.

A worked example: finding the speed

(slide 4)

Bologna to Florence is about 100 km. If we cover it in 1 h, what is the average speed?

1. $v = \frac{s}{t} = \frac{100 \text{ km}}{1 \text{ h}} = 100 \text{ km/h}$: the average speed.
2. $100 \text{ km/h} \div 3.6 \approx 27.8 \text{ m/s}$: convert to SI units.

A worked example: finding the time

(slide 5)

A cyclist rides at 20 km/h. How long does she take to cover 5 km?

Now the unknown is the time, so we rearrange: $t = s/v$.

$$t = \frac{5}{20} = 0.25 \text{ h}$$

and we stop here: this is not yet in minutes.

Before moving on. How many minutes is 0.25 h?

Not 25: an hour has 60 minutes, so $0.25 \times 60 = 15$ minutes.

Try it yourself

(slide 6–7)

Four exercises. The solutions are at the end of the chapter.

1. A car travels 180 km in 2 h. Find v in km/h and in m/s.
2. A runner moves at 4 m/s for 10 min. How many metres does she cover?
3. A train travels at 120 km/h. How long does it take to cover 300 km? Give hours and minutes.
4. Convert 15 m/s into km/h.

Reading a distance–time graph

(slide 8–9)

The same idea can be drawn. On a distance–time graph the **slope** of the line is the speed (figure 1).

- The steeper the line, the faster the motion.
- A flat (horizontal) line means the object is stationary.
- A downward slope means the object is moving back the other way.

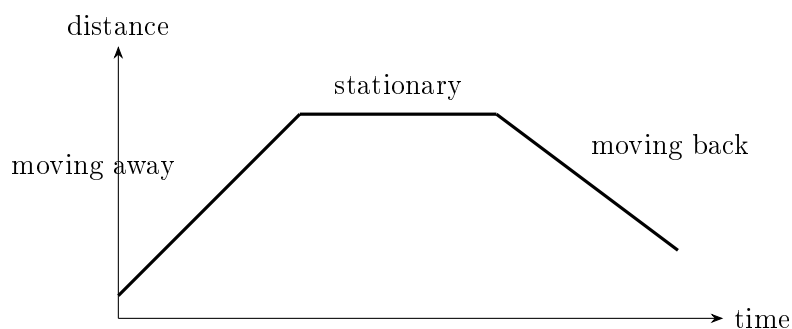


Figure 1: Three stretches of a distance–time graph: a rising line, a horizontal line, a falling line (sketch, no scale).

A past-paper graph

(slide 10)

Here is a graph from a Cambridge IGCSE 0653 past paper (figure 2). The car and the bicycle both start from the same point and travel in the same direction along the same road.

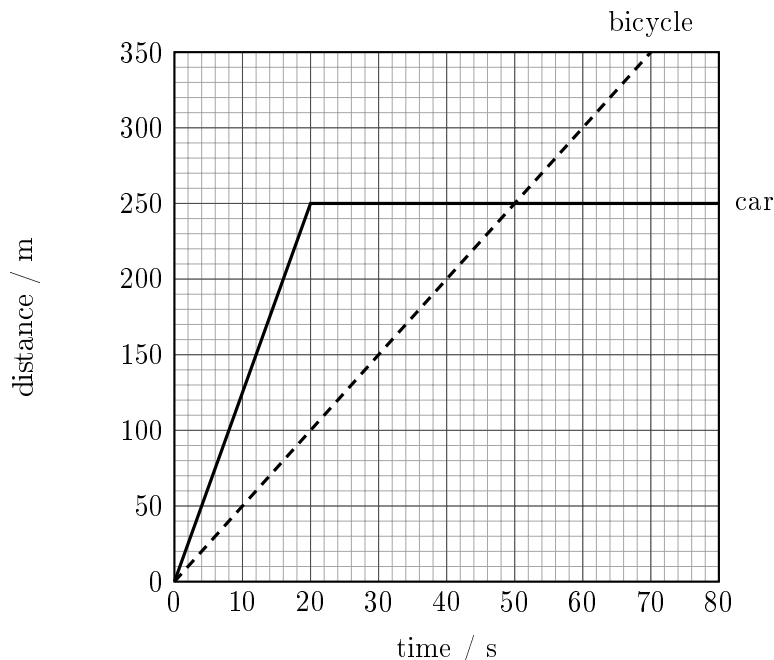


Figure 2: Fig. 9.1 from a Cambridge IGCSE 0653 past paper: distance–time graphs for a car journey and a bicycle journey. Both start from the same point and travel in the same direction along the same road (redrawn).

Before moving on. Use the graph to describe the car journey [2 marks]. At what time does the bicycle pass the car [1 mark]? Only a qualitative reading is needed: no calculation.

The car starts fast: its line is steep, and it covers 250 m in the first 20 s. Then the line becomes horizontal: the car is stationary at 250 m. The bicycle moves along a less steep line that keeps rising, and it catches the car where its line meets the car's horizontal line, at 50 s.

Source: Cambridge IGCSE Combined Science 0653, past paper (Fig. 9.1).

Quick check

(slide 12–13)

A multiple-choice question from a past paper, then a short calculation.

1. A man climbs up a vertical cliff that is 60 m high. He takes two hours to reach the top. What is the average vertical speed of the man?
A 0.0083 m/s **B** 0.50 m/s **C** 30 m/s **D** 120 m/s
2. A cyclist travels 30 km in 2 h. What is her average speed? Answer in km/h.

Source of question 1: Cambridge IGCSE Combined Science 0653, past paper.

Summary

(slide 14)

1. Average speed: $v = s/t$.
2. SI unit m/s; common unit km/h; to convert, divide or multiply by 3.6.
3. On a distance–time graph, slope = speed.

Solutions to the exercises

(slide 6–7, 12–13)

Try it yourself

1. $v = \frac{180}{2} = 90 \text{ km/h}$. In m/s: $90 \div 3.6 = 25 \text{ m/s}$.
2. $t = 10 \text{ min} = 10 \times 60 = 600 \text{ s}$. $s = vt = 4 \times 600 = 2\,400 \text{ m} = 2.4 \text{ km}$.
3. $t = \frac{s}{v} = \frac{300}{120} = 2.5 \text{ h}$. The 0.5 is half an hour: $2.5 \text{ h} = 2 \text{ h } 30 \text{ min}$.
4. $15 \times 3.6 = 54 \text{ km/h}$.

Quick check

1. Convert first: $2 \text{ h} = 2 \times 3600 = 7\,200 \text{ s}$. $v = \frac{60}{7200} \approx 0.0083 \text{ m/s}$. The answer is **A**.
2. $v = \frac{30}{2} = 15 \text{ km/h}$.